

SILICON LABS EMBEDDED SYSTEMS TECHNICAL REFERENCE MANUAL & SPECIFICATION

SiWG917 Dual-Node V2X Telemetry Platform

Hardware Subsystem Orchestration, Dual-Core Tasking, 802.11 UDP Datagram Protocol & Crash Hysteresis Engine

SoC: SIWG917M111MGTBA	Dev Board: BRD2605A (Rev A02)	SDK: Simplicity SDK 2025.12.3 / WiSeConnect 3
Wire Protocol: v2x_frame_t (42 Bytes)	Toolchain: ARM GCC 12.2.rell	Doc Status: Production Engineering Reference

1. EXECUTIVE ARCHITECTURAL OVERVIEW

The SiWG917 V2X Telemetry Node platform is a safety-critical, ultra-low-power embedded communications architecture engineered for cooperative vehicular awareness, emergency deceleration flagging, and telemetric state propagation[cite: 5]. Executing across the Silicon Labs SiWG917 dual-core processor architecture, the firmware decouples high-bandwidth 802.11 b/g/n wireless networking from deterministic physical inertial sensing[cite: 5]. A real-time sensor ingestion engine interfaces with an on-board TDK InvenSense ICM40627 6-axis IMU via an Ultra-Low-Power Synchronous Serial Interface (ULP SSI), running an anti-chatter hysteresis state machine that verifies emergency vehicle deceleration maneuvers within 120 ms[cite: 5].

2. MICROCONTROLLER HARDWARE & PERIPHERAL CLOCK ROUTING

The SIWG917M111MGTBA integrates an ARM Cortex-M4 application processor running up to 180 MHz alongside a network wireless processor (NWP) co-core running isolated proprietary radio firmware[cite: 5]. Hardware peripherals operate within isolated power domains[cite: 5]:

Subsystem	Hardware Domain	Clock Source & Routing	Configured Mode / Frequency	Source Component
Application Core	Base MCU (M4 Core)	M4 PLL / High-Frequency RC	180 MHz active CPU clock	system_si91x.c
ULP SSI Master	ULP Domain (Power State 2)	ULPCLK_GPIO / Primary Divider	10.0 MHz SPI clock, CPOL=0 / CPHA=0	sl_si91x_ssi.c
Sensor Rail Switch	UULP (Ultra-ULP) NPSS	UULP_VBAT_GPIO logic core	NPSS GPIO Pin 1, MUX Mode 1 (Push-Pull)	sl_si91x_driver_gpio.c
Wireless Baseband	NWP Offload Engine	RF Synthesizer / Calibrated XTAL	2.4 GHz Band, 802.11b/g/n (Auto/Ch 6)	sl_wifi.c / sl_net.c
High-Res Timebase	SYSRTC Sleep Timer	32.768 kHz Low-Power Crystal / RO	Continuous millisecond & tick tracking	sl_sleeptimer.c

3. HARDWARE INTERFACING & SENSOR POWER SEQUENCING

The ICM40627 motion sensor resides on a switched power domain to prevent parasitic leakage current through ESD protection diodes during low-power sleep intervals[cite: 5]. Initialization strictly enforces hardware-calibrated settling windows[cite: 5]:

- **Clock Gate:** `sl_si91x_gpio_driver_enable_clock(ULPCLK_GPIO)` [cite: 5].
 - **Mux Selection:** NPSS GPIO Pin 1 mapped to `NPSS_GPIO_PIN_MUX_MODEL` [cite: 5].
 - **Rail Drive:** Direction set to `GPIO_OUTPUT`; driven logic HIGH (1) [cite: 5].
 - **Capacitor Stabilization Window:** Enforced 25 ms settling delay via SYSRTC before initiating SPI transactions [cite: 5].
 - **Bus Stabilization:** Post-SSI init settling delay of 10 ms [cite: 5].
 - **Software Reset:** Sensor register reset followed by 15 ms recovery [cite: 5].
- Hardware Power-On Invariant:** Never issue SPI transactions to the ICM40627 while NPSS GPIO 1 is low [cite: 5]. Doing so will back-feed current through the SPI MISO/MOSI clamping diodes into the unpowered sensor core, causing peripheral latch-up and brown-out states on the ULP power plane [cite: 5].

ULP SSI Bus Characteristics

- **Target Peripheral:** `SL_SSI_ULP_PRIMARY_ACTIVE`, 10.0 MHz master clock [cite: 5].
- **Clock Polarity & Phase:** CPOL=0 (Clock idle low), CPHA=0 (Data latched on first leading edge) [cite: 5].
- **Bit Width & Routing:** 8-bit frame transmissions, Chip Select routed to `SSI_SLAVE_0` [cite: 5].

- **Signature Verification:** Queries register `WHO_AM_I` (0x75) and asserts signature 0x4E[cite: 5].

4. IMU SENSOR ENGINE & RING BUFFER (ACCELEROMETER.C)

The sensor driver maintains an internal, lock-free circular FIFO sample buffer designed for low-overhead telemetry ingestion and historical sequence aggregation[cite: 5]:

Symbol / Attribute	Allocated Value	Memory Footprint	Architectural Role
s_sample_buffer[]	acc_data_t[100]	1,200 Bytes (SRAM)	Continuous ring buffer storing IEEE 754 float vectors (X, Y, Z in g) [cite: 5].
IMU_BUFFER_CAPACITY	100 elements	N/A	Circular depth, accommodating 4.0 seconds of history at 25 Hz ODR[cite: 5].
s_head_index	volatile int	4 Bytes	Points to the next writable slot in the circular array[cite: 5].
s_total_samples	volatile uint32_t	4 Bytes	Monotonically increasing counter of total valid conversions ingested[cite: 5].
ACCELEROMETER_FREQ	25 Hz	N/A	Output Data Rate (ODR = 40 ms per discrete conversion)[cite: 5].
SEQUENCE_LENGTH	90 samples	N/A	Window length required for AI/ML inference readiness check (3.6 sec)[cite: 5].

Sampling & Retrieval Mechanics: `accelerometer_poll()` runs synchronously at the TX schedule deadline, reads conversion registers via SSI, stores the vector in `s_sample_buffer[s_head_index]`, advances the head index modulo 100, and increments total samples[cite: 5]. Instantaneous access is guaranteed via `accelerometer_get_latest()`, which computes $(s_head_index - 1 + CAPACITY) \% CAPACITY$ to return the latest conversion in O(1) time without bus stalling or mutex contention[cite: 5].

5. WIRE PROTOCOL SPECIFICATION: BASIC SAFETY MESSAGE (V2X_FRAME_T)

All over-the-air frames enforce strict byte packing (`#pragma pack(push, 1)`) yielding an exact 42-octet payload footprint[cite: 5]:

Offset (Hex)	Field Identifier	Data Type	Octets	Description & Protocol Constraints
0x00 - 0x03	magic	uint32_t	4 B	Protocol Magic Word: 0x56325831 ("V2X1" in big-endian ASCII)[cite: 5].
0x04	version	uint8_t	1 B	Protocol Format Version: 0x01[cite: 5].
0x05	message_type	uint8_t	1 B	0x01 = Periodic Telemetry, 0x02 = Emergency Braking, 0x03 = Gesture[cite: 5].
0x06 - 0x07	payload_length	uint16_t	2 B	Total length of wire frame: 42 Bytes (<code>sizeof(v2x_frame_t)</code>)[cite: 5].
0x08 - 0x09	sequence	uint16_t	2 B	Monotonically incrementing packet counter for loss and jitter tracking[cite: 5].
0x0A - 0x0F	sender_id[6]	uint8_t[6]	6 B	Hardware IEEE 802.11 MAC address of the transmitting node interface[cite: 5].
0x10 - 0x13	timestamp_ms	uint32_t	4 B	Hardware uptime tick translated to milliseconds via SYSRTC[cite: 5].
0x14 - 0x17	accel_x	float	4 B	Longitudinal acceleration in g units (IEEE 754 float)[cite: 5].
0x18 - 0x1B	accel_y	float	4 B	Lateral acceleration in g units (IEEE 754 float)[cite: 5].
0x1C - 0x1F	accel_z	float	4 B	Vertical acceleration in g units (IEEE 754 float)[cite: 5].
0x20	gesture	uint8_t	1 B	Gesture/ML ID (0: Wing, 1: Ring, 2: Slope, 3: No Gesture)[cite: 5].
0x21	emergency	uint8_t	1 B	Safety Flag: 0 = Normal/Safe, 1 = Hard Deceleration Warning[cite: 5].
0x22 - 0x23	reserved	uint16_t	2 B	Hardware authentication tag / expansion word (cleared to 0x0000)[cite: 5].
0x24 - 0x27	confidence	float	4 B	Confidence score of event detection (0.000 to 1.000)[cite: 5].
0x28 - 0x29	checksum	uint16_t	2 B	Fletcher-16 checksum evaluated over bytes 0x00 to 0x27 (40 bytes)[cite: 5].

6. FLETCHER-16 CHECKSUM & VALIDATION PIPELINE

The integrity algorithm evaluates two accumulators across the 40 payload octets preceding the checksum field[cite: 5]:

```
Let B = {b_0, b_1, ..., b_{N-1}} where N = offsetof(v2x_frame_t, checksum) = 40.
Initialize S_1 = 0, S_2 = 0.
For k = 0 to N-1:
    S_1 = (S_1 + b_k) mod 255
    S_2 = (S_2 + S_1) mod 255
Checksum = (S_2 << 8) | S_1
```

Four-Stage Inbound Validation Pipeline (v2x_validate_frame)

- 1. **Byte Length Gate:** Asserts received datagram length exactly equals sizeof(v2x_frame_t) (42 Bytes)[cite: 5].
- 2. **Magic Signature Gate:** Asserts frame->magic == 0x56325831 to discard unintended port traffic[cite: 5].
- 3. **Protocol Version Gate:** Asserts frame->version == 1 for schema compatibility[cite: 5].
- 4. **Checksum Integrity Gate:** Recomputes Fletcher-16 over the received payload and asserts equivalence[cite: 5].

7. FREERTOS DUAL-TASKING & SOCKET ARCHITECTURE

Networking is decoupled into two independent FreeRTOS tasks to prevent socket starvation[cite: 5]:

Task Identifier	Task Function	Priority Level	Stack Size	Operational Role
TX Network Task	v2x_net_task_entry	osPriorityNormal	3584 Bytes	Manages network bringup, executes 100 ms schedule, applies jitter, broadcasts frames.
RX Ingestion Task	v2x_rx_task_entry	osPriorityBelowNormal	3072 Bytes	Executes blocking recvfrom(), filters local MAC echoes, validates checksums[cite: 5].

Transmit Anti-Collision Jitter: Evaluates a linear shift permutation tick ^ (tick >> 7) ^ (tick << 9) % 21 yielding 0 to 20 ms pseudo-random delay, preventing CSMA/CA RF lock in synchronized clusters[cite: 5].

NWP Socket Caveats: Calling setsockopt(SO_BROADCAST) returns errno = 42 (ENOPROTOOPT) because the NWP offload co-processor natively routes subnet broadcasts without socket option flags[cite: 5]. The firmware logs a warning and continues nominal execution[cite: 5].

8. CRASH SAFETY CLASSIFIER & DECELERATION ENGINE (BSM.C)

The classifier monitors longitudinal acceleration (accel_x) using dual-counter hysteresis to reject road vibration[cite: 5]:

- **Deceleration Threshold:** -0.50 g (flags severe braking / collision deceleration)[cite: 5].
- **Confirmation Window:** Requires 3 consecutive samples below threshold (120 ms at 25 Hz) before triggering emergency state[cite: 5].
- **Release Window:** Requires 5 consecutive samples above threshold (200 ms at 25 Hz) before clearing emergency state[cite: 5].

9. PHYSICAL VCOM VERIFICATION TRACES

```

--- [NODE A: ACCESS POINT (COM11, Serial 440332563)] ---
=====
                        SiWG917 V2X PLATFORM
=====
[APP] Creating network task...
[V2X NET] Initializing Network Subsystem (Soft AP)...
[V2X NET] AP Active: SSID='V2X_AP_NODE', IP=192.168.10.1
[V2X NET] MAC = 08:DD:EB:CF:46:58
[IMU] ICM40627 ID 0x4E verified.
[IMU] Initialized at 25 Hz sampling rate.
[V2X NET] WARNING: SO_BROADCAST failed, errno=42. Continuing.
[V2X NET] UDP socket ready on port 5000.
[V2X RX] RX TASK STARTED (Buffer: 42 bytes)
[V2X TX] OK | Seq=4083 | Bytes=42 | Accel=(0.971, 0.072, 0.211) g | Peer=YES
[V2X TX] OK | Seq=4084 | Bytes=42 | Accel=(0.965, 0.089, 0.245) g | Peer=YES

--- [NODE B: STATION CLIENT (COM21, Serial 440332449)] ---
=====
                        SiWG917 V2X STATION NODE + ICM40627
=====
[V2X NET] Scanning 2.4 GHz channels for AP 'V2X_AP_NODE'...
[V2X NET] Station interface is UP.
[V2X NET] Station MAC = 08:DD:EB:CF:3F:A0
[IMU] ICM40627 ready on Board B.
[V2X NET] UDP socket bound to port 5000.
[V2X NET] Broadcast destination: 192.168.10.255:5000
[V2X RX] RX TASK STARTED
[V2X RX] recvfrom returned: 42 bytes
[V2X RX] VALID PEER PACKET | Seq=4086 | MAC=08:DD:EB:CF:46:58 | Accel=(0.984, 0.071, 0.210) g | Checksum=
0x6100 (MATCH)

```

10. FLASHING & DEPLOYMENT RUNBOOK

```

# 1. Enumerate connected development boards via Simplicity Commander
$ commander adapter list
440332563   BRD2605A SiWG917 (Access Point Node A)
440332449   BRD2605A SiWG917 (Station Client Node B)

# 2. Flash Access Point binary to Node A
$ commander flash v2x_ap_node.rps --device SIWG917M11MGTBA --serialno 440332563

# 3. Flash Station binary to Node B
$ commander flash v2x_station_node.rps --device SIWG917M11MGTBA --serialno 440332449

```